生物统计和生物信息引论

Lecture 8

生存分析方法

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填空题 2分

设置

此题未设置答案，请点击右侧设置按钮

某患者：

- 真实生存时间 T=18
- 随访在 12 月结束

- Y = [填空1] δ = [填空2]

作答

Recall: NCCTG Lung Cancer Data

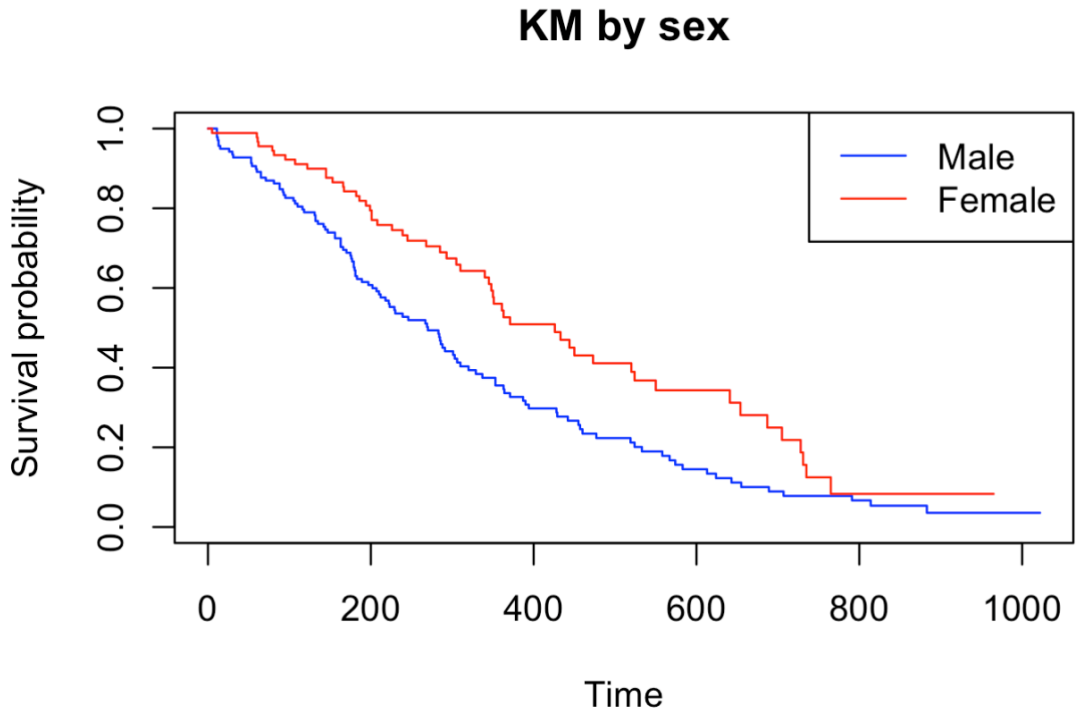
- survival::lung in R
- The North Central Cancer Treatment Group (NCCTG) in the United States.
- **228 patients** with advanced lung cancer.
- Target outcome: Time (survival time, in days)
- Covariates: age, sex, etc
- How to handle covariates?

	inst	time	status	age	sex	ph.ecog	ph.karno	pat.karno	meal.cal	wt.loss
1	3	306	2	74	1	1	90	100	1175	NA
2	3	455	2	68	1	0	90	90	1225	15
3	3	1010	1	56	1	0	90	90	NA	15
4	5	210	2	57	1	1	90	60	1150	11
5	1	883	2	60	1	0	100	90	NA	0
6	12	1022	1	74	1	1	50	80	513	0
7	7	310	2	68	2	2	70	60	384	10
8	11	361	2	71	2	2	60	80	538	1
9	1	218	2	53	1	1	70	80	825	16
10	7	166	2	61	1	2	70	70	271	34
11	6	170	2	57	1	1	80	80	1025	27
12	16	654	2	68	2	2	70	70	NA	23
13	11	728	2	68	2	1	90	90	NA	5
14	21	71	2	60	1	NA	60	70	1225	32
15	12	567	2	57	1	1	80	70	2600	60
16	1	144	2	67	1	1	80	90	NA	15
17	22	613	2	70	1	1	90	100	1150	-5

Recall: Two Group Comparison

- Data: NCCTG Lung Cancer Data
- K-M estimator by sex

```
12  fit_km_group <- survfit(Surv(time, status) ~ sex, data =
    lung, conf.type = "log-log")
13
14  plot(fit_km_group, col = c("blue", "red"), conf.int = FALSE
    , xlab = "Time", ylab = "Survival probability", main = "KM
    by sex")
15
16  legend("topright", legend = c("Male", "Female"), col = c
    ("blue", "red"), lty = 1)
```



Example: Log-Rank Test

Log-Rank test: if there is statistical difference between male and female.

```
log_rank_test <- survdiff(Surv(time, status) ~ sex, data = lung)

print(log_rank_test)
```

```
Call:
survdiff(formula = Surv(time, status) ~ sex, data = lung)

        N Observed Expected (O-E)^2/E (O-E)^2/V
sex=1 138      112     91.6      4.55      10.3
sex=2  90       53     73.4      5.68      10.3

 Chisq= 10.3  on 1 degrees of freedom, p= 0.001
```

Limitation of the Log-Rank Test

KM estimator

	inst	time	status	age	sex	ph.ecog	ph.karno	pat.karno	meal.cal	wt.loss
1	3	306	2	74	1	1	90	100	1175	NA
2	3	455	2	68	1	0	90	90	1225	15
3	3	1010	1	56	1	0	90	90	NA	15
4	5	210	2	57	1	1	90	60	1150	11
5	1	883	2	60	1	0	100	90	NA	0
6	12	1022	1	74	1	1	50	80	513	0
7	7	310	2	68	2	2	70	60	384	10
8	11	361	2	71	2	2	60	80	538	1
9	1	218	2	53	1	1	70	80	825	16
10	7	166	2	61	1	2	70	70	271	34
11	6	170	2	57	1	1	80	80	1025	27
12	16	654	2	68	2	2	70	70	NA	23
13	11	728	2	68	2	1	90	90	NA	5
14	21	71	2	60	1	NA	60	70	1225	32
15	12	567	2	57	1	1	80	70	2600	60
16	1	144	2	67	1	1	80	90	NA	15
17	22	613	2	70	1	1	90	100	1150	-5

What if we want to evaluate a continuous variable? (e.g., Age)

What if we need to adjust for multiple covariates simultaneously? (e.g., Age + Sex + ECOG score)

Can the Kaplan-Meier method handle this?

No

单选题 1分

设置

关于Kaplan-Meier（K-M）估计量，以下哪项陈述是正确的?

A K-M估计量可直接用于拟合含连续协变量（如年龄）的回归模型

B K-M估计量是非参数方法，无需对生存时间分布作任何假设

C K-M曲线交叉说明两组生存差异具有统计学意义

D K-M估计量本质上是参数模型，其基线风险函数 $h_0(t)$ 被假定为指数分布

提交

Parametric vs Nonparametric

- Parametric model: a fixed functional form with finite parameters
  - Distribution assumption on data: for example, normal distribution, with 2 parameters (mean and variance)
  - Parametric method: linear regression, logistic regression, likelihood-based method, etc
  - High efficiency, low flexibility; handles covariates well
- Nonparametric model: allow the data to determine the function shape with potentially infinite-dimensional parameters
  - No parametric assumption on data.
  - Nonparametric method: empirical CDF, Wilcoxon rank test, Kruskal-Wallis test, K-M estimator, etc
  - High flexibility, low efficiency; struggles with high dimensional covariates

Methods in Survival Analysis

- Nonparametric method: no assumptions on survival time T.
  - K-M estimation, N-A estimation.

- Parametric method: distribution assumption on T
  - For example: exponential distribution
  - Why not normal distribution?
  - A natural limitation on T: $T \geq 0$

Exponential Distribution

Exponential distribution is the **simplest** parametric distribution for survival time. When $T \sim \text{Exp}(\lambda)$:

- $f(t) = \lambda \exp(-\lambda t)$
- $F(t) = 1 - \exp(-\lambda t)$
- $S(t) = \exp(-\lambda t)$
- $\lambda(t) = \lambda$
- $\Lambda(t) = \lambda t$
- Memoryless property: $\Pr(T > t) = \Pr(T > t + s \mid T > s)$ for $s > 0$
- $\mu = \mathbb{E}(T) = 1/\lambda$, $\text{var}(T) = 1/\lambda^2$

Likelihood-Based Inference

When censoring is **absent**, the likelihood function is

$$L(\lambda; \boldsymbol{t}) = \prod_{i=1}^{n} f(t_i \mid \lambda).$$

The estimator for λ is the solution to

$$\frac{\partial\{\log L(\lambda)\}}{\partial \lambda} = 0.$$

When censoring is **present**, things are different:

For censored individuals, we only know that the survival time is larger than the observed time.

Likelihood for TTE Data

Basic setup. For each subject $i = 1, \ldots, n$, we observe (T_i, δ_i), where T_i is the observed time and δ_i is the event indicator. Assume **independent censoring**.

- Case 1: $\delta_i = 1$. T_i is the survival time.

  Likelihood contribution: $f(T_i)$

  (The probability that the survival time is **exactly** T_i)

- Case 2: $\delta_i = 0$. T_i is the censoring time. We only know the survival time is larger than T_i.

  Likelihood contribution: $S(T_i)$.

  (The probability that the survival time is **larger than** T_i)

Total likelihood contribution: $\{f(T_i)\}^{\delta_i}\{S(T_i)\}^{1-\delta_i}$.

Likelihood for Time-to-event Data

Basic setup. For each subject $i = 1, \ldots, n$, we observe (T_i, δ_i), where T_i is the observed time and δ_i is the event indicator. Assume **independent censoring**.

Likelihood contribution of single observation. $\{f(T_i)\}^{\delta_i}\{S(T_i)\}^{1-\delta_i}$.

Full likelihood.

$$L = \prod_{i=1}^{n}[\{f(T_i)\}^{\delta_i}\{S(T_i)\}^{1-\delta_i}] = \prod_{i=1}^{n}[\{h(T_i)\}^{\delta_i}S(T_i)].$$

$$l = \sum_{i=1}^{n}[\delta_i \log\{h(T_i)\} + \log\{S(T_i)\}].$$

Inference for Exponential Distribution

Likelihood: $L(\lambda) = \prod_{i=1}^{n} \{f(T_i)\}^{\delta_i} \{S(T_i)\}^{1-\delta_i} = \prod_{i=1}^{n} (\lambda \mathrm{e}^{-\lambda T_i})^{\delta_1} (\mathrm{e}^{-\lambda T_i})^{1-\delta_i}.$

Log likelihood: $l(\lambda) = (\log \lambda) \sum_{i=1}^{n} \delta_i - \lambda \sum_{i=1}^{n} T_i.$

Estimation: $\frac{\partial l(\lambda)}{\partial \lambda} = 0 \quad \Rightarrow \quad \hat{\lambda} = \frac{\sum_{i=1}^{n} \delta_i}{\sum_{i=1}^{n} T_i}.$

Fisher information: $I(\lambda) = -\frac{\partial^2 l(\lambda)}{\partial \lambda^2} = \frac{\sum_{i=1}^{n} \delta_i}{\lambda^2}.$

As a result: $\hat{\lambda} \xrightarrow{d} \mathcal{N}\left(\lambda, \frac{\lambda^2}{\sum_{i=1}^{n} \delta_i}\right).$

Since $\lambda > 0,$ a better choice: (via delta method)

$$\log(\hat{\lambda}) \approx \mathcal{N}\left(\log \lambda, \frac{1}{\sum_{i=1}^{n} \delta_i}\right).$$

NCCTG Lung Cancer Data

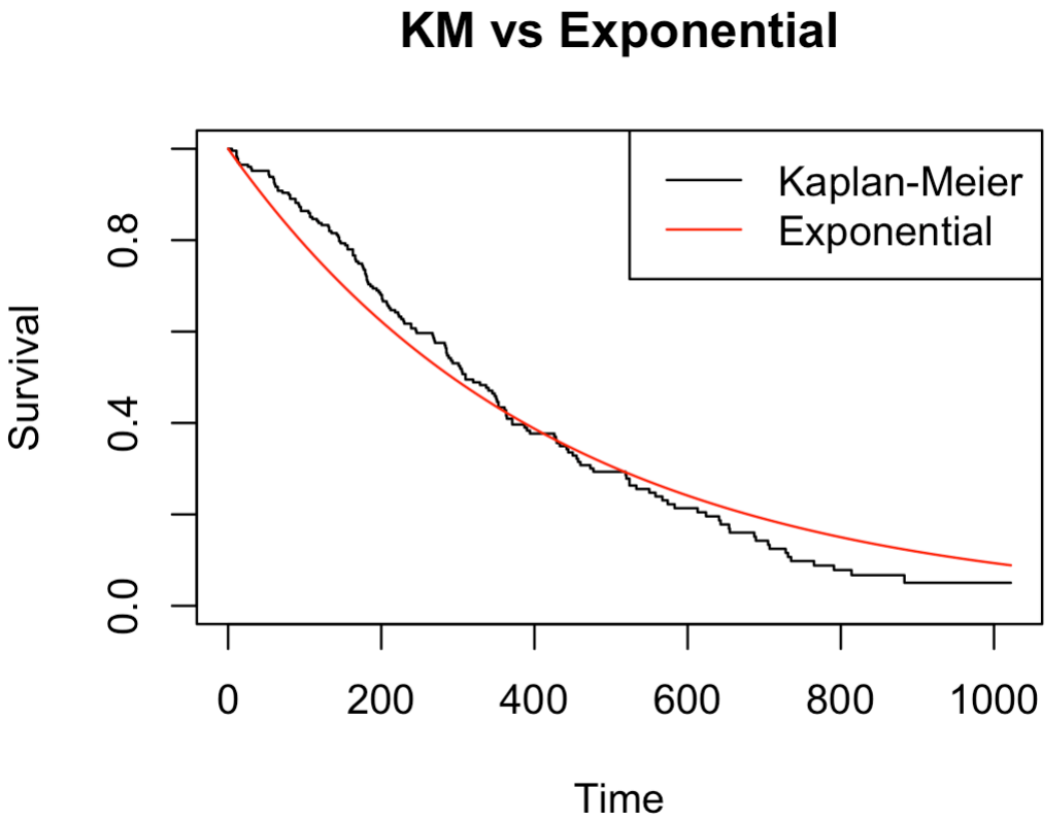

```r
# fit exponential distribution
fit_exp <- survreg(Surv(time, status) ~ 1, dist = "exponential",
data = lung)
# lambda_hat <- sum(lung$status - 1) / sum(lung$time)
lambda_hat <- exp(-coef(fit_exp)[1])

# fit KM
fit_km <- survfit(Surv(time, status) ~ 1, data = lung)

# plot KM
plot(fit_km, conf.int = FALSE,
     xlab = "Time", ylab = "Survival",
     main = "KM vs Exponential")

# plot exponential
curve(exp(-lambda_hat * x), add = TRUE, col = "red")

legend("topright",
       legend = c("Kaplan-Meier", "Exponential"),
       col = c("black", "red"), lwd = 1)
```



Extension: Weibull Distribution

Weibull distribution $W(p, \lambda)$ with scale parameter $\lambda > 0$ and shape parameter $p > 0$.

When $p = 1$, $W(1, \lambda) = \text{Exp}(\lambda)$. Properties of $W(p, \lambda)$:

- $S(t) = \exp\{-(\lambda t)^p\}$
- $f(t) = p\lambda(\lambda t)^{p-1} \exp\{-(\lambda t)^p\}$
- $\lambda(t) = p\lambda(\lambda t)^{p-1}$
- $\Lambda(t) = (\lambda t)^p$

Extension: Gamma Distribution

Gamma distribution $\Gamma(\alpha, \lambda)$. When $\alpha = 1$, $\Gamma(1, \lambda) = \mathrm{Exp}(\lambda)$.

1. The density function

$$f(t) = \frac{\lambda^\alpha t^{\alpha-1} \exp(-\lambda t)}{\Gamma(\alpha)} \propto t^{\alpha-1} \exp(-\lambda t),$$

   where $\Gamma(\alpha) = \int_0^\infty t^{\alpha-1} \exp(-t) \mathrm{d}t$. For integer α, $\Gamma(\alpha) = (\alpha - 1)!$.

2. There is no close formula for survival or hazard function.

3. $\mathbb{E}(T) = \alpha\lambda^{-1}$, $\mathrm{var}(T) = \alpha\lambda^{-2}$.

4. If $T_i \sim \Gamma(\alpha_i, \lambda)$ are independent, then $\sum_{i=1}^K T_i \sim \Gamma(\sum_{i=1}^K \alpha_i, \lambda)$.

NCCTG Lung Cancer Data


```r
# fit KM
fit_km <- survfit(Surv(time, status) ~ 1, data = lung)

# plot KM
plot(fit_km, conf.int = FALSE,
     xlab = "Time", ylab = "Survival",
     main = "KM vs Exponential vs Weibull")

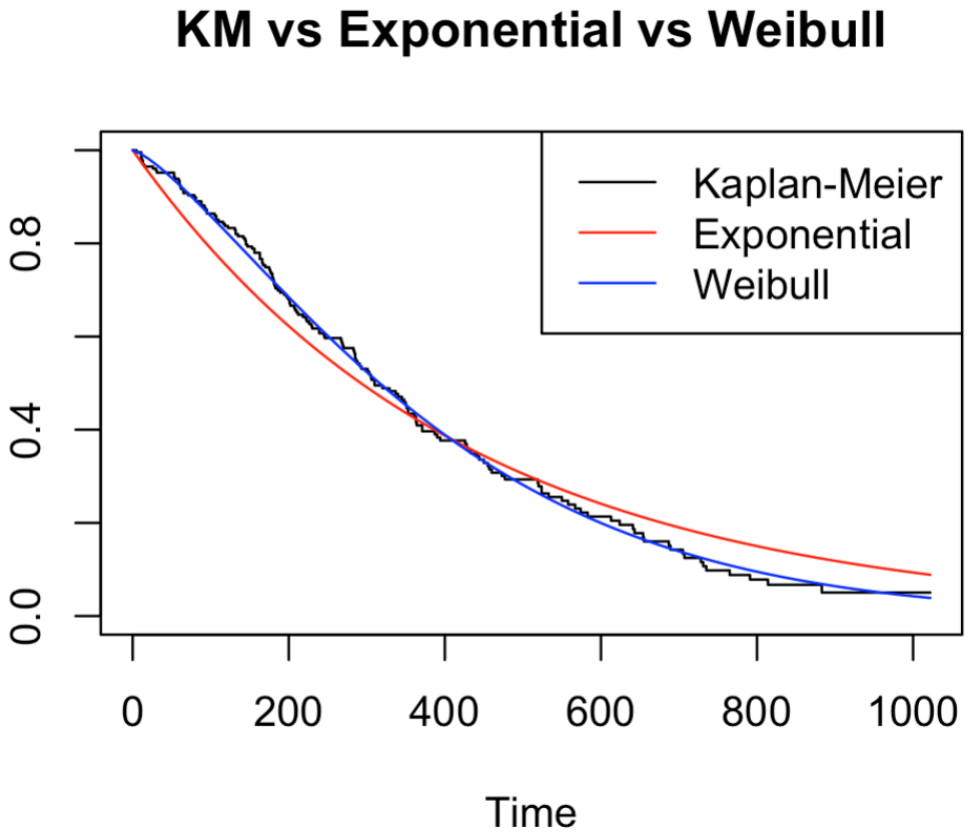
# fit exponential distribution
fit_exp <- survreg(Surv(time, status) ~ 1, dist = "exponential", data = lung)
# lambda_hat <- sum(lung$status - 1) / sum(lung$time)
lambda_hat <- exp(-coef(fit_exp)[1])

# fit Weibull distribution
fit_weib <- survreg(Surv(time, status) ~ 1, dist = "weibull", data = lung)
mu_weib <- coef(fit_weib)[1]
sigma_weib <- fit_weib$scale
shape_weib <- 1 / sigma_weib
scale_weib <- exp(mu_weib)

# plot exponential
curve(exp(-lambda_hat * x), add = TRUE, col = "red")

# plot weibull
curve(exp(-(x / scale_weib)^shape_weib), add = TRUE, col = "blue")

legend("topright",
       legend = c("Kaplan-Meier", "Exponential", "Weibull"),
       col = c("black", "red", "blue"), lwd = 1)
```



How to Handle Covariates?

Recall: hazard function

$$\lambda(t) = \lim_{h \to 0} \frac{\Pr(t \le T < t + h \mid T \ge t)}{h} = \frac{f(t)}{S(t)}.$$

A way to incorporate covariates:

$$\lambda(t \mid \boldsymbol{Z}) = \lambda_0(t) \cdot g(\boldsymbol{Z}).$$

Considering $\lambda(t \mid \boldsymbol{Z}) > 0$ for all t, one choice of $g(\boldsymbol{Z})$ is

$$g(\boldsymbol{Z}) = \exp(\boldsymbol{\beta}^\top \boldsymbol{Z}).$$

Question: Why not directly model observed time T?

Cox Proportional Hazards Model

$$\lambda(t \mid \boldsymbol{Z}) = \lambda_0(t) \cdot \exp(\boldsymbol{\beta}^\top \boldsymbol{Z}).$$

- Semiparametric model
  - It combines a non-parametric part (time) with a parametric part (covariates).
- Breaking down the components:
- Z: The vector of covariates.
  - Can handle continuous (age), discrete (sex), and interactions (age x sex).
- $\lambda_0(t)$: The baseline hazard function.
  - Think of it as the underlying risk for a "baseline" patient where all $Z_i = 0$.

$$\lambda(t \mid \boldsymbol{Z} = \mathbf{0}) = \lambda_0(t) \cdot \exp(\beta_1 \times 0 + \beta_2 \times 0 + \cdots + \beta_p \times 0) = \lambda_0(t).$$

Separating Time and Covariates

Cox proportional hazards model:

$$\lambda_i(t) = \lambda(t \mid \boldsymbol{Z}_i) = \underbrace{\lambda_0(t)}_{\substack{\text{Function of } t, \\ \text{but not } \boldsymbol{Z}}} \cdot \underbrace{\exp(\boldsymbol{Z}_i^\top \boldsymbol{\beta})}_{\substack{\text{Function of } \boldsymbol{Z}, \\ \text{but not } t}} .$$

The hazard function for each subject has **the same general shape** $\lambda_0(t)$, but can be, say, doubled or halved depending on the patient's risk factors (covariates).

Question: Should the model have an intercept in the linear combination $\boldsymbol{Z}^\top \boldsymbol{\beta}$?

No. It can be included in $\lambda_0(t)$.

What does "Proportional" Mean?

Binary covariate case: $Z_i \in \{0, 1\}$.

- For example: $Z_i = 1$ treatment; $Z_i = 0$ control.
- $\lambda_1(t)$: the hazard rate for treated group.
- $\lambda_0(t)$: the hazard rate for control group.

$$\lambda_1(t) = \lambda(t \mid Z = 1) = \lambda_0(t) \exp(\beta).$$

- Hazard Ratio (HR):

$$\frac{\lambda_1(t)}{\lambda_0(t)} = \exp(\beta)$$

constant, not depend on time t

Generalizing the Hazard Ratio

For any two covariates $\boldsymbol{z}_0$ and $\boldsymbol{z}_1$,

$$\text{HR} = \frac{\lambda(t \mid \boldsymbol{z}_1)}{\lambda(t \mid \boldsymbol{z}_0)} = \frac{\lambda_0(t)\exp(\boldsymbol{z}_1^\top \boldsymbol{\beta})}{\lambda_0(t)\exp(\boldsymbol{z}_0^\top \boldsymbol{\beta})} = \exp\{(\boldsymbol{z}_1 - \boldsymbol{z}_0)^\top \boldsymbol{\beta}\}.$$

- HR remains constant over time.
- In general, $\boldsymbol{\beta} = \mathbf{0} \Leftrightarrow \text{HR} = 1 \Leftrightarrow \boldsymbol{Z}$ is not associated with T.
- $\text{HR} = 1$: no effect.
- $\text{HR} > 1$: increased hazard. (e.g., a risk factor like smoking)
- $\text{HR} < 1$: decreased hazard. (e.g., a protective factor like a successful treatment)

Inference under Cox Model

Recall: full likelihood for TTE data

$$L(\lambda) = \prod_{i=1}^{n} \{f(T_i \mid \boldsymbol{Z}_i)\}^{\delta_i} \{S(T_i \mid \boldsymbol{Z}_i)\}^{1-\delta_i}.$$

- Under the Cox model, both f and S depend on the baseline hazard λ_0.
- Inference requires joint estimation of $\boldsymbol{\beta}$ and λ_0.
- However, λ_0 is difficult to handle, since it may be **infinite-dimensional**.
- One possible approach: impose a model on λ_0.
- Alternative approach: partial likelihood, which **bypass** direct modeling of λ_0.

Partial Likelihood

Consider the ordered event times $t_1 < t_2 < \cdots < t_m$, where t_k corresponds to unit $i_k \in \{1, 2, \ldots, n\}$. Note that $m \leq n$ due to censoring.

Alternative approach for constructing likelihood function:

$$L(\boldsymbol{\beta}) = \prod_{k=1}^{m} \Pr\{i_k \text{ fails at } t_k \mid R(t_k)\},$$

where $R(t_k) = \{j : T_j \geq t_k\}$ is the risk set at t_k.

Under the Cox model, the failure probability is a **weighted** probability over the risk set:

$$\Pr\{i_k \text{ fails at } t_k \mid R(t_k)\} = \frac{\lambda(t_k \mid \boldsymbol{Z}_{i_k})}{\sum_{j \in R(t_k)} \lambda(t_k \mid \boldsymbol{Z}_j)} = \frac{\lambda_0(t) \exp(\boldsymbol{Z}_{i_k}^\top \boldsymbol{\beta})}{\sum_{j \in R(t_k)} \lambda_0(t) \exp(\boldsymbol{Z}_j^\top \boldsymbol{\beta})} = \frac{\exp(\boldsymbol{Z}_{i_k}^\top \boldsymbol{\beta})}{\sum_{j \in R(t_k)} \exp(\boldsymbol{Z}_j^\top \boldsymbol{\beta})}.$$

Partial likelihood:

$$L(\boldsymbol{\beta}) = \prod_{k=1}^{m} \frac{\exp(\boldsymbol{Z}_{i_k}^\top \boldsymbol{\beta})}{\sum_{j \in R(t_k)} \exp(\boldsymbol{Z}_j^\top \boldsymbol{\beta})}.$$

only consider event time points!

Partial Likelihood

Equivalent expression of partial likelihood:

$$L(\boldsymbol{\beta}) = \prod_{i=1}^{n} \left\{ \frac{\exp(\boldsymbol{Z}_i^\top \boldsymbol{\beta})}{\sum_{j \in R(T_i)} \exp(\boldsymbol{Z}_j^\top \boldsymbol{\beta})} \right\}^{\delta_i}.$$

Exercise: write the partial likelihood for the following survival data.

- Remission time (in weeks) for 17 leukemia patients.
- $Z = 1$ treatment; $Z = 0$ control.

ID	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
T	6	6+	7	9+	10	10+	11+	13	16	17+	19+	20+	22	25+	32+	34+	35+
Z	1	0	1	0	1	1	0	0	0	1	1	0	1	0	0	1	1

Partial Likelihood

ID	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
T	6	6+	7	9+	10	10+	11+	13	16	17+	19+	20+	22	25+	32+	34+	35+
Z	1	0	1	0	1	1	0	0	0	1	1	0	1	0	0	1	1

$$L(\beta) = \frac{\exp(\beta Z_1)}{\sum_{j=1}^{17}\exp(\beta Z_j)} \cdot \frac{\exp(\beta Z_3)}{\sum_{j=3}^{17}\exp(\beta Z_j)} \cdot \frac{\exp(\beta Z_5)}{\sum_{j=5}^{17}\exp(\beta Z_j)} \cdot \frac{\exp(\beta Z_8)}{\sum_{j=8}^{17}\exp(\beta Z_j)} \cdot \frac{\exp(\beta Z_9)}{\sum_{j=9}^{17}\exp(\beta Z_j)} \cdot \frac{\exp(\beta Z_{13})}{\sum_{j=13}^{17}\exp(\beta Z_j)}$$

$$= \frac{\exp(\beta)}{9\exp(\beta)+8} \cdot \frac{\exp(\beta)}{8\exp(\beta)+7} \cdot \frac{\exp(\beta)}{7\exp(\beta)+6} \cdot \frac{1}{5\exp(\beta)+5} \cdot \frac{1}{5\exp(\beta)+4} \cdot \frac{\exp(\beta)}{3\exp(\beta)+2}.$$

Estimation and Inference

Partial likelihood:

$$L(\boldsymbol{\beta}) = \prod_{i=1}^{n} \left\{ \frac{\exp(\boldsymbol{Z}_i^\top \boldsymbol{\beta})}{\sum_{j \in R(T_i)} \exp(\boldsymbol{Z}_j^\top \boldsymbol{\beta})} \right\}^{\delta_i}.$$

Estimator of $\boldsymbol{\beta}$:

$$\hat{\boldsymbol{\beta}} = \arg\max_{\boldsymbol{\beta}} \{\log L(\boldsymbol{\beta})\}.$$

Asymptotic dirtsibution:

$$\hat{\boldsymbol{\beta}} \approx \mathcal{N}(\boldsymbol{\beta}, I^{-1}(\boldsymbol{\beta})), \quad I(\boldsymbol{\beta}) = -\frac{\partial^2 \{\log L(\boldsymbol{\beta})\}}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}^\top}.$$

How to Handle Ties in Partial Likelihood

- Recall: partial likelihood requires $t_1 < t_2 < \ldots < t_m$.
- Question: Hwo to handle ties, i.e., multiple events occcurring at the same time?
- Breslow approximation (simplest)
  - Treat tied events as occurring simultaneously.

$$\frac{\prod_{i \in D(t)} \exp(\boldsymbol{Z}_i^\top \boldsymbol{\beta})}{\left\{\sum_{j \in R(t)} \exp(\boldsymbol{Z}_j^\top \boldsymbol{\beta})\right\}^{|D(t)|}},$$

  where $D(t)$ is the set of individuals who fail at time t.
- Efron approximation (default in practice, also default in R)
  - Approximates ties as occurring sequentially within a small interval.

$$\frac{\prod_{i \in D(t)} \exp(\boldsymbol{Z}_i^\top \boldsymbol{\beta})}{\prod_{l=0}^{|D(t)|-1} \left\{\sum_{j \in R(t)} \exp(\boldsymbol{Z}_j^\top \boldsymbol{\beta}) - \frac{l}{d} \sum_{i \in D(t)} \exp(\boldsymbol{Z}_i^\top \boldsymbol{\beta})\right\}}.$$

NCCTG Lung Cancer Data


```
35 # fit cox model
36 fit_cox <- coxph(Surv(time, status) ~ age + sex, data = lung)
37 summary(fit_cox)
```

```
> summary(fit_cox)
Call:
coxph(formula = Surv(time, status) ~ age + sex, data = lung)

  n= 228, number of events= 165

        coef exp(coef)  se(coef)      z Pr(>|z|)
age  0.017045  1.017191  0.009223  1.848  0.06459 .
sex -0.513219  0.598566  0.167458 -3.065  0.00218 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

    exp(coef) exp(-coef) lower .95 upper .95
age    1.0172     0.9831    0.9990    1.0357
sex    0.5986     1.6707    0.4311    0.8311

Concordance= 0.603  (se = 0.025 )
Likelihood ratio test= 14.12  on 2 df,   p=9e-04
Wald test            = 13.47  on 2 df,   p=0.001
Score (logrank) test = 13.72  on 2 df,   p=0.001
```

Proportional Hazard Assumption

Main assumption: **Proportional hazard assumption**.

$$\frac{\lambda(t \mid \boldsymbol{Z}_1)}{\lambda(t \mid \boldsymbol{Z}_2)} = \exp\{(\boldsymbol{Z}_1 - \boldsymbol{Z}_2)^\top \boldsymbol{\beta}\} \qquad \textbf{not depend on } t.$$

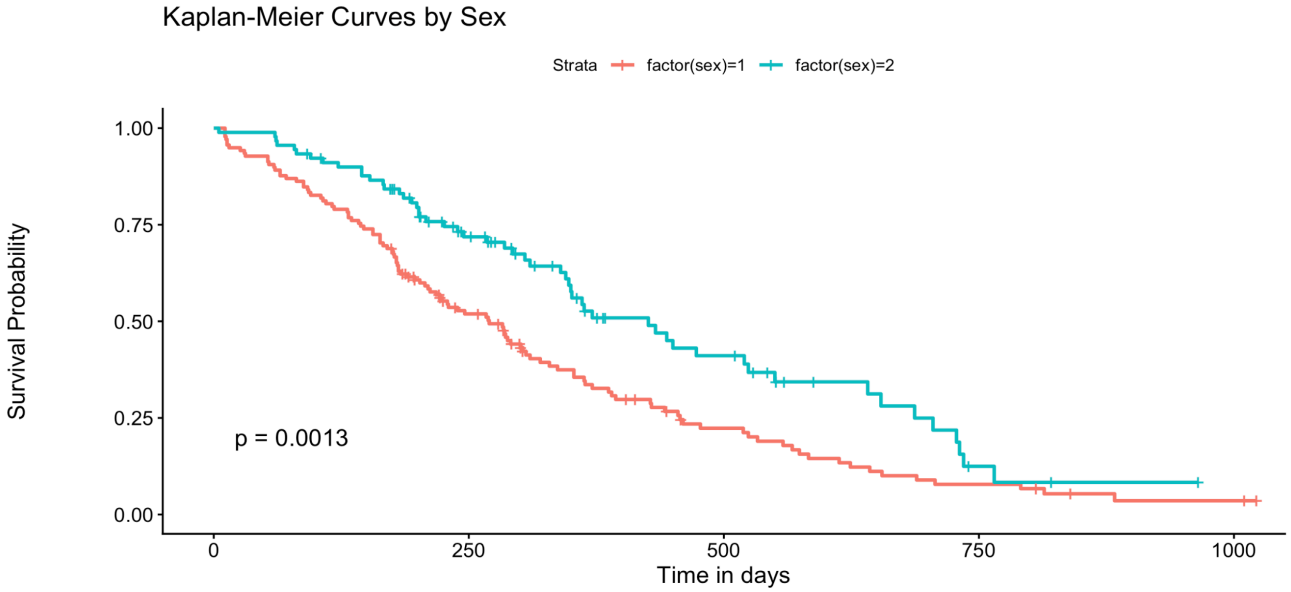
How to check this assumption?

1. $\log(-\log S(t))$ plot. Under the assumption, $S(t \mid \boldsymbol{Z}) = \{S_0(t)\}^{\exp(\boldsymbol{Z}^\top \boldsymbol{\beta})}$.
   $\log(-\log S(t \mid \boldsymbol{Z})) = \log(-\log S_0(t)) + \boldsymbol{Z}^\top \boldsymbol{\beta}$.
   Plot $\log(-\log \hat{S}(t))$ vs. $\log(t)$, where $\hat{S}$ is the Kaplan–Meier estimation.
2. Schoenfeld residual test: test if residuals are independent of time.
   `cox.zph` in `R`.

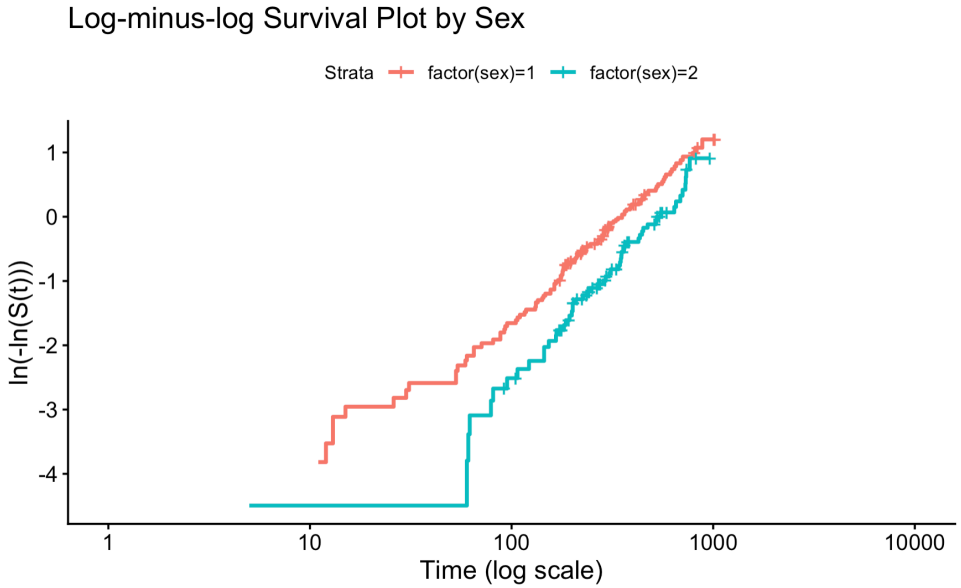
Other assumption: independent censoring, linear structure.

Proportional Hazard Assumption

Kaplan-Meier curves: If the survival curves **cross**, the assumption is likely violated!

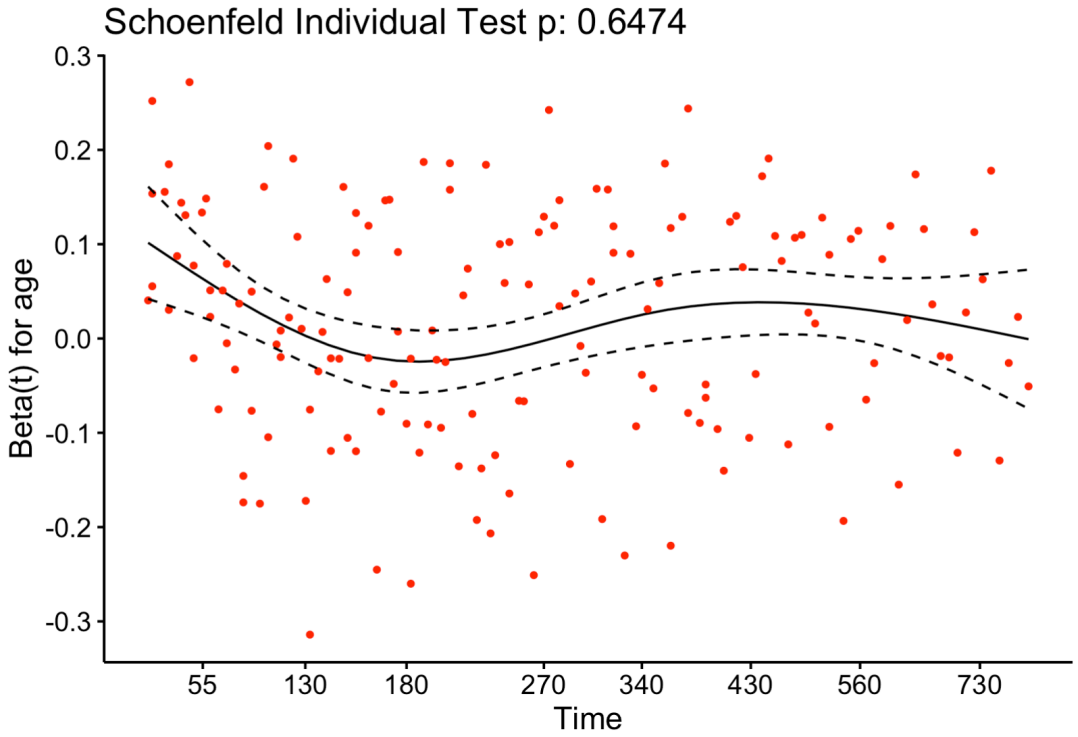


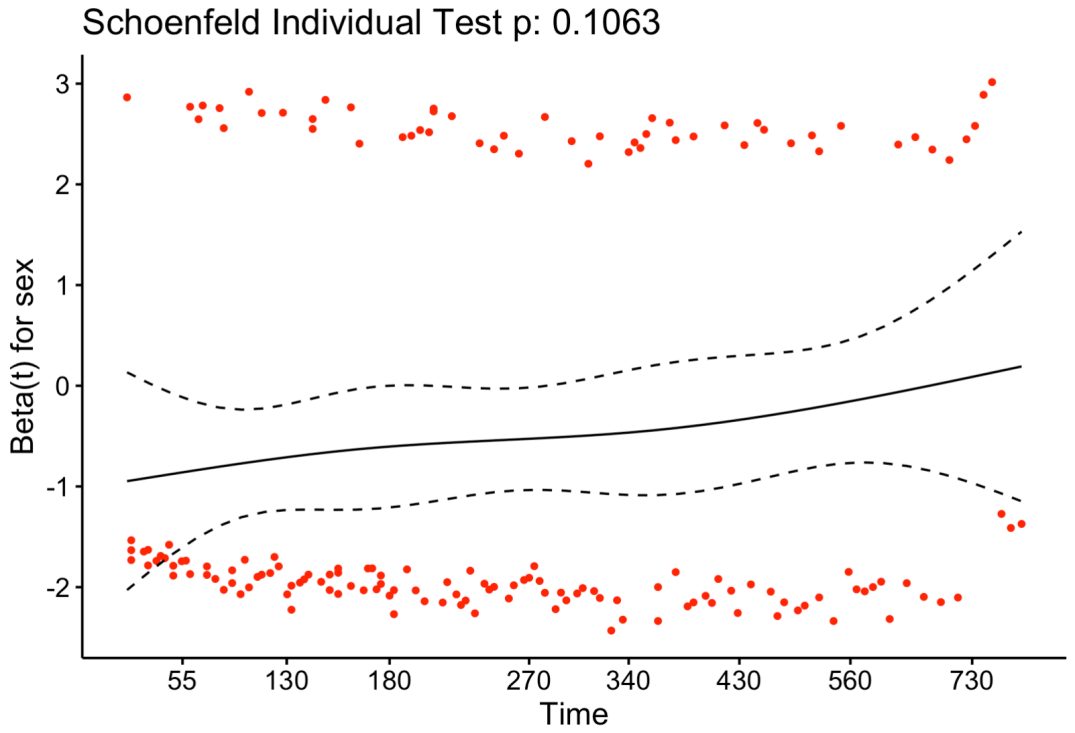
Log-minus-log survival plot: Plotting $\log(-\log S(t))$ vs. $\log(t)$.



Schoenfeld Residual Test


```
35  # fix cox model
36  fit_cox <- coxph(Surv(time, status) ~ age + sex, data = lung)
37  summary(fit_cox)
38
39  # Schoenfeld
40  cox.zph(fit_cox)
41  survminer::ggcoxzph(cox.zph(fit_cox))
```





Accelerated Failure Time (AFT) Model

Recall: Cox proportional hazards model.

$$\lambda(t \mid \boldsymbol{Z}) = \lambda_0(t) \cdot g(\boldsymbol{Z}).$$

Question: Why not directly model observed time T?
For example:

$$T = T_0 \cdot g(\boldsymbol{Z}).$$

Considering $T > 0$, one choice of $g(\boldsymbol{Z})$ is

$$g(\boldsymbol{Z}) = \exp(\boldsymbol{\beta}^\top \boldsymbol{Z} + \varepsilon).$$

Accelerated Failure Time (AFT) Model

Accelerated Failure Time (AFT) Model.

$$T = T_0 \exp(\boldsymbol{Z}^\top \boldsymbol{\beta} + \varepsilon),$$

where $\varepsilon_i \overset{\text{i.i.d.}}{\sim} f(\varepsilon)$.

Interpretation:

- Longer survival when $\boldsymbol{Z}_i^\top \boldsymbol{\beta} + \varepsilon_i > 0$ (Deceleration)
- Shorter survival when $\boldsymbol{Z}_i^\top \boldsymbol{\beta} + \varepsilon_i < 0$ (Acceleration)

If life is a burning candle, the Cox model measures the size of the flame, while the AFT model measures how fast the candle burns.

Interpretation of AFT Model

Log-linear expression of AFT model:

$$\log T_i = \mu + \boldsymbol{Z}_i^\top \boldsymbol{\beta} + \sigma W_i.$$

- $\mu = \log T_0$ is the intercept.
- $\sigma > 0$ is the scale parameter.
- W_i is assumed to follow a distribution (e.g., normal).
- $\varepsilon_i = \sigma W_i$ is the error term.

Unknown parameters: μ, $\boldsymbol{\beta}$ and σ.

Inference for AFT Model

Log-linear expression of AFT model:

$$\log T_i = \mu + \boldsymbol{Z}_i^\top \boldsymbol{\beta} + \sigma W_i.$$

Unknown parameters: $\boldsymbol{\beta}$ and σ.
Likelihood:

$$f_T(t_i|z_i) = \frac{1}{\sigma t_i} f_W\left(\frac{\log t_i - \mu - \boldsymbol{z}_i^\top \boldsymbol{\beta}}{\sigma}\right)$$

$$L = \prod_{i=1}^{n} \{f_T(t_i)\}^{\delta_i} \{S_T(t_i)\}^{1-\delta_i}$$

$$\propto \prod_{i=1}^{n} \left\{\frac{1}{\sigma} f_W\left(\frac{\log t_i - \mu - \boldsymbol{z}_i^\top \boldsymbol{\beta}}{\sigma}\right)\right\}^{\delta_i} \left\{S_W\left(\frac{\log t_i - \mu - \boldsymbol{z}_i^\top \boldsymbol{\beta}}{\sigma}\right)\right\}^{1-\delta_i}.$$

Inference can be conducted using likelihood theory.

Distribution of W

The remaining part is to specify the distribution of W_i:

- When $W_i \sim \mathcal{N}(0,1)$, T follows a log-normal distribution. (log of normal)
- When the distribution function of W_i is $F_W(w) = 1 - \exp\{-\exp(w)\}$:

$$\Pr(T > t \mid \boldsymbol{Z}) = \Pr\left(W > \frac{\log t - \mu - \boldsymbol{Z}^\top \boldsymbol{\beta}}{\sigma}\right) = \exp\left\{-\exp\left(\frac{\log t - \mu - \boldsymbol{Z}^\top \boldsymbol{\beta}}{\sigma}\right)\right\}$$

$$= \exp\left\{-t^{1/\sigma} \exp\left(-\frac{\mu + \boldsymbol{Z}^\top \boldsymbol{\beta}}{\sigma}\right)\right\}.$$

Recall: Weibull distribution $S(t) = \exp(-(\gamma t)^p)$.

$T \mid \boldsymbol{Z} \sim$ Weibull(p, γ) with $\gamma = \exp(-\mu - \boldsymbol{Z}^\top \boldsymbol{\beta})$ and $p = 1/\sigma$.

NCCTG Lung Cancer Data


```
43 # AFT with weibull
44 fit_AFT <- survreg(Surv(time, status) ~ age + sex, data = lung, dist
   = "weibull")
45 summary(fit_AFT)
```

```
> summary(fit_AFT)

Call:
survreg(formula = Surv(time, status) ~ age + sex, data = lung,
    dist = "weibull")
               Value Std. Error     z       p
(Intercept)  6.27485    0.48137 13.04 < 2e-16
age         -0.01226    0.00696 -1.76  0.0781
sex          0.38209    0.12748  3.00  0.0027
Log(scale)  -0.28230    0.06188 -4.56 5.1e-06

Scale= 0.754

Weibull distribution
Loglik(model)= -1147.1   Loglik(intercept only)= -1153.9
        Chisq= 13.59 on 2 degrees of freedom, p= 0.0011
Number of Newton-Raphson Iterations: 5
n= 228
```

Verification of the Distribution Assumption

Main assumption: $W_i \sim f_W$.

- When $W_i \sim \mathcal{N}(0,1)$, this assumption can be verified via Q-Q plot of the fitted residuals.
- When the distribution function of W_i is $F_W(w) = 1 - \exp\{-\exp(w)\}$, we have

$$\log(-\log S(t)) = p \log t + C.$$

This assumption can be verified via plotting $\log(-\log \hat{S}(t))$ vs $\log t$.

AFT vs Cox PH

AFT model: $T = T_0 \cdot \exp(\boldsymbol{Z}^\top \boldsymbol{\beta})$.

- $S_T(t \mid \boldsymbol{Z}) = \Pr(T > t \mid \boldsymbol{Z}) = \Pr\{T_0 > t \exp(-\boldsymbol{Z}^\top \boldsymbol{\beta}) \mid \boldsymbol{Z}\} = S_{T_0}\{t \exp(-\boldsymbol{Z}^\top \boldsymbol{\beta})\}$.
- $f_T(t \mid \boldsymbol{Z}) = f_0\{t \exp(-\boldsymbol{Z}^\top \boldsymbol{\beta})\} \cdot \exp(-\boldsymbol{Z}^\top \boldsymbol{\beta})$. (Chain rule)

It gives

$$\lambda(t \mid \boldsymbol{Z}) = \frac{f_T(t \mid \boldsymbol{Z})}{S_T(t \mid \boldsymbol{Z})} = \lambda_0\{t \exp(-\boldsymbol{Z}^\top \boldsymbol{\beta})\} \cdot \exp(-\boldsymbol{Z}^\top \boldsymbol{\beta}).$$

AFT vs Cox PH

- Cox PH model: $\lambda(t \mid \boldsymbol{Z}) = \lambda_0(t) \cdot \exp(\boldsymbol{Z}^\top \boldsymbol{\alpha})$.

$$\log \lambda(t \mid \boldsymbol{Z}) = \log \lambda_0\{\exp(\log t)\} + \boldsymbol{Z}^\top \boldsymbol{\alpha}.$$

- AFT model: $\lambda(t \mid \boldsymbol{Z}) = \lambda_0\{t \exp(-\boldsymbol{Z}^\top \boldsymbol{\beta})\} \cdot \exp(-\boldsymbol{Z}^\top \boldsymbol{\beta})$.

$$\begin{aligned}\log \lambda(t \mid \boldsymbol{Z}) &= \log \lambda_0\{\exp(\log t - \boldsymbol{Z}^\top \boldsymbol{\beta})\} - \boldsymbol{Z}^\top \boldsymbol{\beta} \\ &= \log \lambda_{\text{base}} + \log \lambda_0\{\exp(\log t - \boldsymbol{Z}^\top \boldsymbol{\beta})\} - \log \lambda_0\{\exp(\log t)\},\end{aligned}$$

where $\log \lambda_{\text{base}} = \log \lambda_0\{\exp(\log t)\} - \boldsymbol{Z}^\top \boldsymbol{\beta}$.

- On the $\log \lambda$ vs $\log t$ scale
  - PH assumption: to model the change in hazard as a vertical shift. $(\log \lambda)$
  - AFT assumption: to model the change in hazard as a horizontal shift. $(\log t)$

AFT vs Cox PH

- Cox PH model: $\lambda(t \mid \boldsymbol{Z}) = \lambda_0(t) \cdot \exp(\boldsymbol{Z}^\top \boldsymbol{\alpha})$.
- AFT model: $\lambda(t \mid \boldsymbol{Z}) = \lambda_0\{t \exp(-\boldsymbol{Z}^\top \boldsymbol{\beta})\} \cdot \exp(-\boldsymbol{Z}^\top \boldsymbol{\beta})$.
- The Weibull distribution is the only distribution that satisfies both the Cox PH and AFT formulations.
  - $\lambda_0(t) = p\gamma(\gamma t)^{p-1}$.
  - Cox PH: $\lambda(t) = p\gamma(\gamma t)^{p-1} \exp(\boldsymbol{Z}^\top \boldsymbol{\alpha})$.
  - $S(t) = \exp\{-\int_0^t \lambda(u)\mathrm{d}u\} = \exp\{-\gamma^p t^p \exp(\boldsymbol{Z}^\top \boldsymbol{\alpha})\}$.
  - AFT: $S(t) = \exp[-t^{1/\sigma} \exp\{-(\mu + \boldsymbol{Z}^\top \boldsymbol{\beta})/\sigma\}]$.
- Correlations: $p = 1/\sigma$, $\gamma^p = \exp(-\mu/\sigma)$, $\boldsymbol{\alpha} = -\boldsymbol{\beta}/\sigma$.

Example: Whole-genome sequencing (WGS)

Provides a tumor-agnostic solution for detecting an increasingly complex range of DNA-based biomarkers.

WGS succeeded in **89%** of cases (793 out of 888 samples) with a median turnaround time of about 6 working days.

nature medicine

Article

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Real-world clinical utility of tumor whole-genome sequencing in solid cancers

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Check for updates

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Molecular testing is essential in precision oncology. Whole-genome sequencing (WGS) provides a tumor-agnostic solution for detecting an increasingly complex range of DNA-based biomarkers. Here we present real-world data from 888 patients to demonstrate the clinical utility of routine, paired tumor-normal WGS diagnostics for solid cancers in a comprehensive cancer center. WGS succeeded in 89% of cases with a median turnaround time of 6 working days. Potentially actionable biomarkers were identified in 73% of patients, including biomarkers for reimbursed (27%) and experimental (63%) therapies. Within 1 year, 40% and 19% of patients, respectively, started biomarker-informed treatment, which was associated with a 31% longer median overall survival (+96 days) compared with patients not receiving such therapy. Among patients without prior systemic therapy, biomarker-informed treatment yielded significantly longer overall survival (median not reached) than non-biomarker-informed therapy (427 days) or no systemic therapy (214 days). In cancers of unknown primary ($n = 123$), WGS contributed to diagnostic solution or detected biomarker-driven reimbursed treatment options in 67%, with 68% starting tumor-type-specific therapy. Clinically relevant pathogenic germline variants were identified in 6.5% of patients. Overall, WGS-based diagnostics had clinical consequences for 41% of tested patients, providing a versatile tool for routine clinical practice in solid oncology.

Example: data colletion

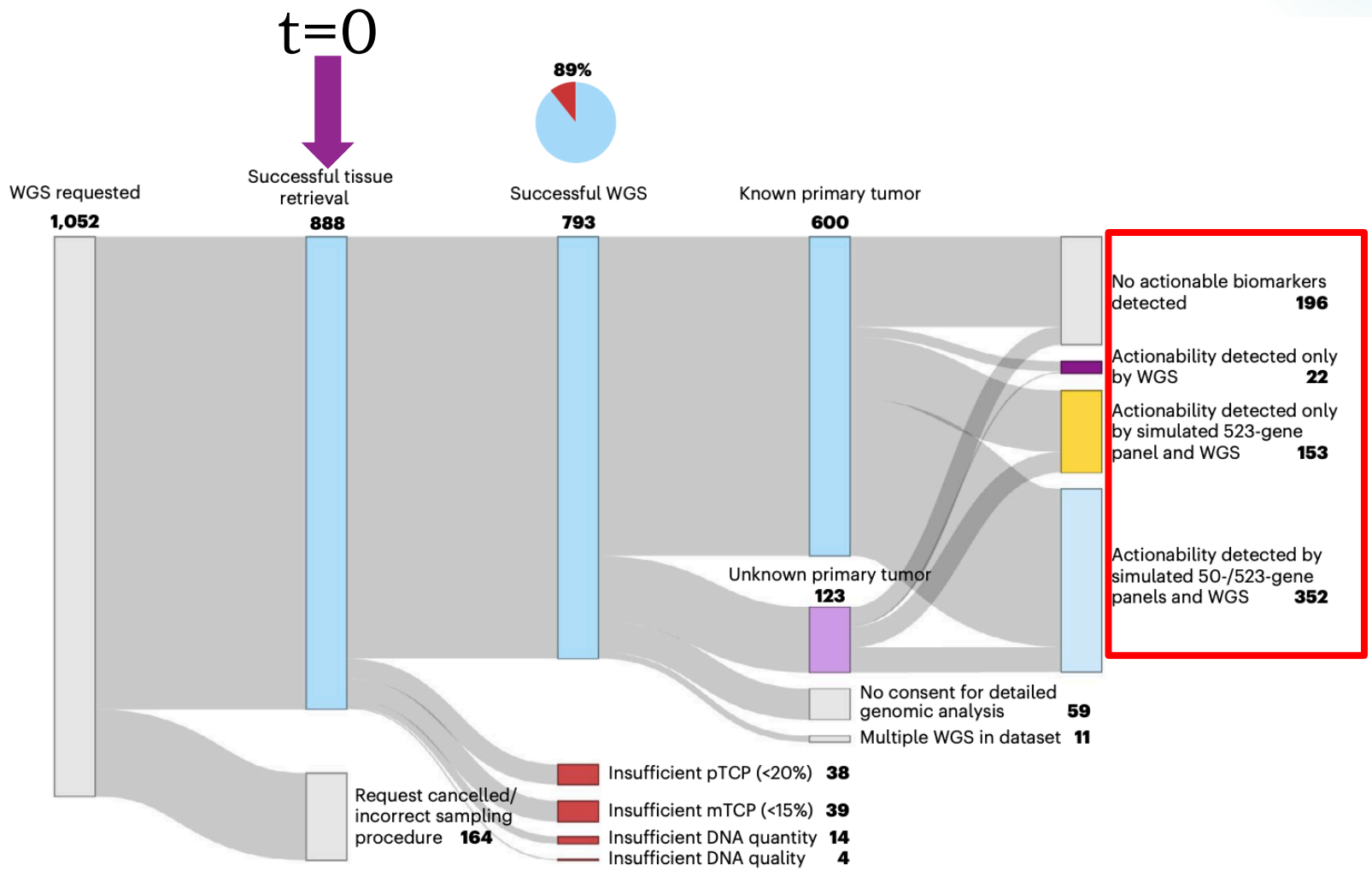
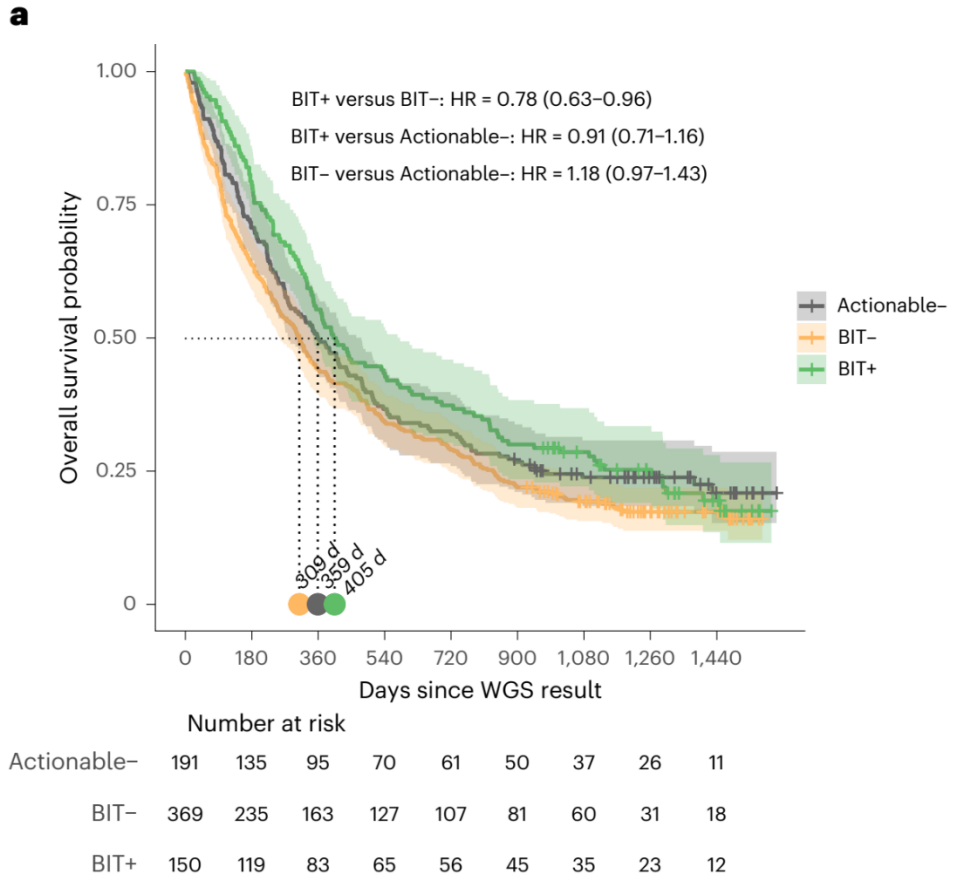


Fig. 1 | Feasibility of WGS in routine clinical practice and simulated actionable biomarker coverage by small and comprehensive sequencing panels. Starting from 1,052 total WGS requests, reasons for discontinuation and failure as well as diagnostics outcomes are indicated. Patients who did not have consent for patient-level genomic data analysis were included only for general patient characteristics. Panel coverage of a commonly used 50-gene panel and a 523-gene panel was simulated. Potential actionability includes standard-of-care and experimental biomarker-informed treatments. pTCP, pathologist-assessed tumor cell percentage.

Broad Target Identification: Potentially actionable biomarkers were identified in **73%** of patients. (527 out of 723)

- **27%** of patients had biomarkers for reimbursed standard-of-care therapies.
- **63%** of patients had biomarkers for experimental or clinical trial therapies

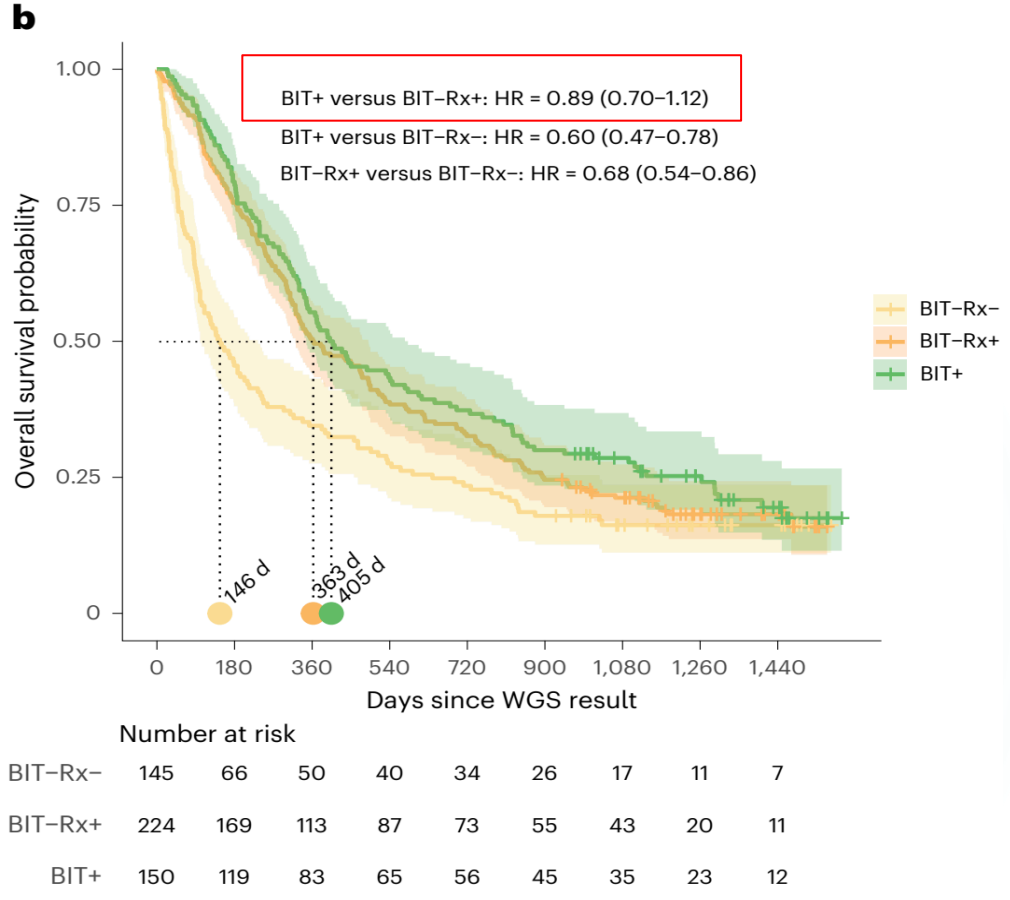
Example: Whole-genome sequencing (WGS)



a, (Actionable−)patients without potentially actionable biomarkers patients with ⩾1.
(BIT−)potentially actionable biomarker who did not receive a biomarker-informed treatment after WGS patients with ⩾1.
(BIT+)potentially actionable biomarker who did receive biomarker-informed therapy.

WGS-Informed Therapy & Overall Survival (OS)
•**Significant Survival Benefit**: Biomarker-informed treatment (BIT+) guided by WGS led to a **31% increase** in median Overall Survival (+96 days) compared to patients without BIT.

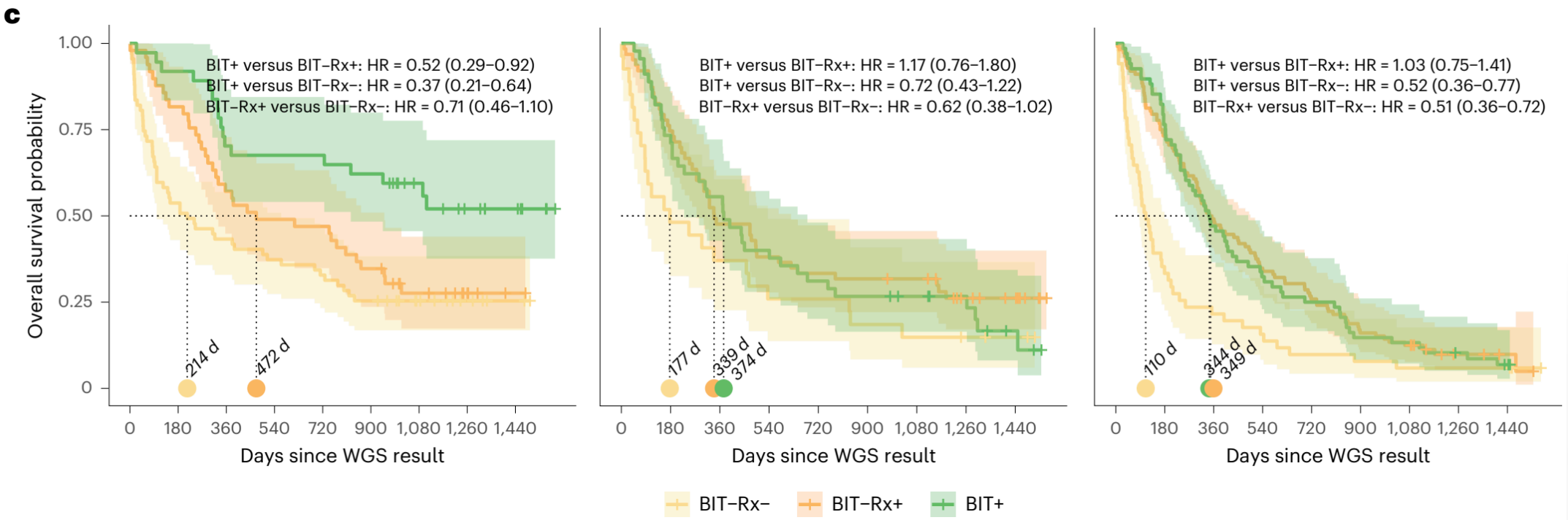
Example: Whole-genome sequencing (WGS)



b, Within the actionable cohort, patients not receiving biomarker-informed therapy are split into

- those receiving only non-biomarker-informed systemic therapy (BIT−Rx+) and
- those without any post-WGS systemic therapy (BIT−Rx−), shown alongside patients who received biomarker-informed treatment (BIT+).

•**Superiority Over Standard Care**: BIT+ (median OS: 405 days) ~~significantly~~ outperformed non-biomarker systemic therapy (363 days) and no therapy (146 days).



c, Among patients with ⩾1 potentially actionable biomarker, OS is stratified by the number of prior systemic therapy lines at the time of WGS: 0 (left,初期), 1 (center, 中期) or ⩾2 (right, 后期).
Dotted lines indicate the median survival and are annotated with the corresponding number of days. HRs are reported with 95% CIs in parentheses.

- **The "Timing" Effect**:
  - **Maximal Benefit in First-Line:** The OS advantage was most pronounced in **treatment-naive patients** (0 prior lines), where median OS for BIT+ was not yet reached after 4 years.
  - **Diminishing Returns Late-Line:** No significant survival advantage was observed when BIT+ was initiated after multiple prior lines of systemic therapy.
- **Clinical Takeaway**: WGS should be implemented **early in the diagnostic pathway** to maximize therapeutic impact and patient outcomes

Summary

- Parametric vs Nonparametric
- Likelihood-based inference for TTE data
- Exponential, Weibull, Gamma distribution
- Cox proportional hazards (Cox PH) model
  - Partial likelihood
  - Proportional hazard assumption
  - Schoenfeld residual test
- Accelerated failure time (AFT) model
  - Distribution assumption
- AFT vs Cox PH